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Correlation between insertion torque value and bone density in D3 and D4 bone following photobiomodulation: a randomised interventional study

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Abstract

Background This study aimed to assess the correlation between insertion torque values and bone density measured via CBCT in D3 and D4 bone types after photobiomodulation (PBM) therapy.

Materials and methods 108 partially edentulous patients were divided into six groups based on PBM therapy type and bone classification. Insertion torque values were recorded during implant placement, and bone density was assessed via CBCT.

Results Statistically significant positive correlations were observed between bone density and insertion torque in all regions (cervical, middle, apical) for both D3 and D4 bone types (Spearman's rho ≥ 0.951, $p < 0.01$).

Conclusion Bone density correlates strongly with insertion torque values in both D3 and D4 bone following PBM therapy, suggesting its potential predictive value for implant primary stability.

Clinical significance With the findings of the study, CBCT can be used as a predictive tool for assessing the implant stability and also advocates the use of PBM as an adjunct to improve primary implant stability.

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Keywords Bone density, CBCT, Implant stability, Torque, Photobiomodulation

Introduction

One important issue to consider when forecasting implant stability is bone density. Because of the characteristics of the area, clinical investigations indicate that implant survival is higher in the mandible than in the upper maxilla. Bone quality, or bone density, limits this survival. Implants put in type IV bone had a higher failure risk, according to several studies in the literature. Similarly, we saw favorable outcomes when implants were positioned in type I, II, and III bone, the latter of which is the best kind of bone for sufficient implant stability [1]. Attaining good primary stability is very critical to achieve osseointegration and long-term success. Primary stability is determined primarily by bone density and quality, both of which can be preoperatively evaluated using cone-beam computed tomography (CBCT) [2–4]. Implant stability can be predicted subjectively by surgeons assumption, objectively by insertion torque test

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and Resonance frequency analysis(RFA) after implant placement [5]. A torque gauge included into the drilling unit determines the Insertion torque test; in contrast, an electronic device and a transducer screwed to the implant measures the RFA [6]. CBCT helps in evaluation of bone density which offers clinicians a predictive method for estimating primary stability and potential implant success [7]. A crucial component of preimplant clinical and radiographic examinations is the assessment of bone tissue density, and the advent of Cone Beam Computer Tomography (CBCT) marks a substantial advancement in the application of computed tomography. While Norton and Gamble proposed classifying bone density based on CT images, where the bone quality data at the future implant site are obtained based on objective and quantitative results expressed in Hounsfield units (HU), Misch, Lekholm, and Zarb classified bone density based on the subjective assessment of the patients. HU unit represents a qualitative radiolucency measure of different tissues at CT. HU scale ranges from – 1000 (air), 0 (water) to + 1000 (bone), where the value of this unit depends on the tissue density through which the X rays pass [8]. Hounsfield units can be utilized as a diagnostic metric to evaluate potential implant stability since they have greater bone density (HU) values and a higher primary implant stability measured in ISQ units [9]. Insertion torque, as an indirect indicator of bone quality, has been correlated with resonance frequency analysis (RFA) and other mechanical stability measures [10–12].The BITS (Bone-Insertion Torque-Stability) score further quantifies this interrelationship and is gaining attention for clinical decision-making [13].

Emerging therapeutic interventions such as Photobiomodulation (PBM) have shown promise in enhancing osseointegration, improving crestal bone density, and shortening healing periods [14, 15]. In PBM, "there is no significant increase in target tissue temperature above normal tissue temperature (37°C)" as suggested by its operational definition. The non- thermal curative effects of PBM are different from the heat-mediated ablation caused by laser which are used for high power surgical applications and photodynamic therapy. Lasers at various wavelengths that are popularly used for PBM are "visible (660 nm)", "near-infrared (810 nm and 940 nm)", and less often, "midinfrared (1,040 nm, 2,940 nm, 9,400 nm, or 10,400 nm)". Broad light sources and LED are also becoming famous due to improved performance, excellent quality and control that is offered by the latest devices [16]. It has been demonstrated that using PBM in the early postoperative phase increases the mechanical strength of the bone–implant interface and promotes the formation of bone matrix [17]. Because low-level lasers have the rare capacity to thoroughly permeate tissues and produce positive effects, their use has grown in popularity. LLLT promotes angiogenesis, increases cellular metabolism, and modifies inflammatory responses, all of which improve tissue healing and dental implant osseointegration [18]. PBM works by stimulating mitochondrial activity and bone remodeling processes, making it a potential adjunctive treatment in cases of low-density bone such as D3 and D4 types [19–21].

In literature, there are numerous studies on the usefulness of CT in terms of the assessment of the bone volume and morphology as well as several clinical studies on the relationship between CT values and primary implant stability. However, there has not been a sufficient number of clinical studies attempting to determine the correlation between bone density and the value of primary implant stability.

The purpose of this study was to investigate the relationship between insertion torque values and bone density as determined by CBCT in D3 and D4 which is low density bone after following different PBM therapies, thereby evaluating the utility of PBM in improving primary implant stability.

Materials and methods

A prospective randomised interventional study to estimate the correlation between insertion torque value and bone density using CBCT in D3 and D4 bone after undergoing Photobiomodulation was studied [15]. 108 partially edentulous patients with D3 and D4 bone type undergoing implant placement were chosen from the Prosthodontics Department in D3 or D4 bone were chosen. Written informed consent was given to all patients prior to their involvement in the trial. The study design is Randomised, placebo controlled, double blinded trial with CTRI registration number as CTRI/2022/04/042033 and IEC approval DMIMS(DU)/IEC/2021/628 conducted for a duration of 2 years from June 2022 to June 2024. The commensurate sample size obtained was 108 where we calculated the sample size based on the difference in Mean gray scale values as a primary variable between pre and post treatment intervention control vs. experimental and considering std. dev. Based on values of the reference study by Matys J et al. 2019, the minimum sample size for each group was 18. From the Department of Prosthodontics and Crown & Bridge, the study population are males and females in age group 18 to 70 years were taken based on following inclusion and exclusion criteria.

Inclusion criteria

1. Patients who have given written informed consent.
2. Healthy men and women patients of age 18–70 years.
3. Requisite volume of bone for placement of implant such that the patient doesn't have the need for augmentation of bone.

4. Patients with single missing teeth and indicated for implant therapy for replacement of same having D3 and D4 Type bone quality (150 to 850 Hounsfield units).
5. Patients with multiple missing teeth and implant scheduled at varied time.

Exclusion criteria

6. Patients with systemic disease.
7. Patients with immunocompromised state.
8. Patient with parafunctional habits like bruxism.
9. Patients with present or past history of deleterious habits (smoking/ tobacco/ alcohol).
10. Patients with uncontrolled periodontal disease.
11. Patients who had undergone radiotherapy/ bisphosphonate medication.

They were sub grouped into:

The Groups 1, 2 and 3 belong to D3 bone category which had 2 intervention groups and 1 control group. The Group 1 and 2 were given Photobiomodulation using Therapy A and Therapy B respectively and Group 3 was control group. Similarly, The Groups 4, 5 and 6 belong to D4 bone category which had 2 intervention groups and 1 control group. The Group 4 and 5 were given Photobiomodulation using Therapy A and Therapy B respectively and Group 6 was control group as specified in Table 1.

Group 1- experimental-18 patients PBM Therapy A in D3 bone
Group 2- experimental-18 patients PBM Therapy B in D3 bone
Group 3-control group- 18 patients PBM Therapy C in D3 bone
Group 4- experimental-18 patients PBM therapy A in D4 bone
Group 5- experimental-18 patients PBM Therapy B in D4 bone
Group 6- control group- 18 patients PBM Therapy C in D4 bone.

CBCT and clinical examination was carried out for intended site of implant placement and Osstem implants were placed in D3 and D4 bone according to the groups classified. All surgical operations were carried out by a surgeon and following implant placement, the experimental groups were randomly assigned to undergo the prescribed PBM therapy. The randomisation was done using computer generated randomisation table. The laser irradiation treatment was carried out by a specialist who was blinded to the patient's bone type and group to which he is allocated so that he is not aware of the type of bone. Along with the irradiation specialist, the evaluator was blinded who assessed the bone density changes using CBCT. The PBM therapy was given using Biolase Epic X dental diode laser at a wavelength of 940 nm, Fluence 8 J/cm^2, mode continuous, Buccal and palatal site, in contact with mucosa for 40 s at each side with 4 J each side for all the experimental groups for PBM therapies but the number of laser appointments for PBM therapy A was 7(surgery day, at 2, 4, 6, 8, 10, and 12 days postoperatively) and for PBM therapy B was 5(surgery day, 3, 7, 10, and 14 days postoperatively and PBM therapy was either 7 or 5 days [15]. The insertion torque value was measured using torque wrench after implant placement and the primary stability was assessed using CBCT at baseline after implant surgery. The bone density in Hounsfield units of implant site was measured using CBCT at three different locations viz. apical middle and cervical for all the patients. The measurement was done by dividing

Table 1 Photobiomodulation therapy A, B and C

PBM Specifications	PBM Therapy A	PBM Therapy B	PBM therapy C(Placebo group which will follow either PBM Therapy A or B but with the laser hand piece in kept in off mode)
Output Power	100mW	100mW	NA
Spot area	0.5cm^2	0.5cm^2	NA
Average power density (irradiance)	200 W/cm^2 Or 20mW/cm^2	200 W/cm^2 Or 20mW/cm^2	NA
Fluence	8 J/cm^2	8 J/cm^2	NA
Mode	Continuous	Continuous	NA
Site	Buccal and palatal site, in contact with mucosa	Buccal and palatal site, in contact with mucosa	Buccal and palatal site in contact with mucosa
Time	40 s at each side	40 s at each side	40 s at each side
Energy	4 J each side	4 J each side	NA
Sessions	7(surgery day, at 2, 4, 6, 8, 10, and 12 days postoperatively)	5(surgery day, 3, 7, 10, and 14 days postoperatively	Either A or B
Final dose	54 J	40 J	NA



Fig. 1 CBCT analysis

the whole edentulous area or implant into 3 equal parts i.e. apical, middle and cervical and measuring the value on a straight line that passes through each area and taking out the mean value (Fig. 1) [24]. The correlation was calculated for the insertion torque value and the CBCT measurement.

Results

All the results calculated using SPSS software version 24.0. The results obtained are as follows: For D3 bone, Group 1 the average bone density value was 672.02 after implant placement and the average insertion torque or primary stability value was 40.88, for Group 2 707.8 average bone density and 42.5 is average insertion torque, for Group 3 613.6 is average bone density and 36.3 is average insertion torque. For D4 bone, Group 4 477.7 is average bone density and 28.7 is average insertion torque value, Group 5 489.3 is average bone density and 30.3 is average insertion torque value and for Group 6, 407.4 is average bone density and 26.5 is average insertion torque. Descriptive' and 'analytical' statistics was done. The normality of data was tested by "Kolmogorov-Smirnov" test and was found to be not normal and hence non parametric tests were used. After collaging the data and collecting the values of insertion torque value and bone density across two bone types (D3 and D4), Spearman's correlation coefficient was used to analyse the results and scatter plot was drawn to identify trends like positive and negative correlations. For bone type D3 (Table 2), Cervical shows a very strong positive correlation with a Spearman's rho of 0.99 and a P-value of less than 0.01 (Graph 1), indicating statistical significance. Similarly, Middle and Apical also exhibit strong correlations with Spearman's rho values of 0.989 and 0.963 (Graph 2 and 3), respectively, both with P-values less than 0.01. For bone type D4 (Table 1), Cervical shows a Spearman's rho of 0.951 (Graph 4), and Middle shows a correlation of 0.98 (Graph 5), both indicating strong positive relationships, with P-values less than 0.01. Apical also maintains a strong correlation with a Spearman's rho of 0.96 and a P-value of less than 0.01 (Graph 6). Across all groups, the correlations are significant, demonstrating a consistent and strong relationship between average bone density and insertion torque value.

Table 2 Spearman's correlation results between average bone density and insertion torque values across two bone types (D3 and D4)

Bone Type	Group	Variables	Spearman's rho	*P*-value
D3	Cervical	Average bone density vs. Insertion torque value	0.99	< 0.01
	Middle	Average bone density vs. Insertion torque value	0.989	< 0.01
	Apical	Average bone density vs. Insertion torque value	0.963	< 0.01
D4	Cervical	Average bone density vs. Insertion torque value	0.951	< 0.01
	Middle	Average bone density vs. Insertion torque value	0.98	< 0.01
	Apical	Average bone density vs. Insertion torque value	0.96	< 0.01

Discussion

This study demonstrated a strong and statistically significant correlation between bone density measured via cone-beam computed tomography (CBCT) and insertion torque values across both D3 and D4 bone types. These results are aligned with the growing body of evidence supporting the predictive role of bone density in determining primary implant stability [2–4, 7].

CBCT has emerged as a reliable, non-invasive imaging modality capable of providing high-resolution data regarding the bone morphology and density around potential implant sites. This enables clinicians to make informed decisions before surgical intervention [5, 21]. Our findings further validate CBCT's clinical utility by showing that bone density values in cervical, middle, and apical regions of the implant site significantly correlate with insertion torque values, indicating a proportional relationship between denser bone and improved mechanical anchorage.

The relationship between bone quality and insertion torque has been previously substantiated in both in vitro and in vivo studies. Mikic M et al. concluded that analysis of bone density from CBCT images in the software of the apparatus expressed in HU units can be used to predict the degree of primary stability of self-tapping and non-self-tapping dental implants in bones of densities D1 and D2 on a material of animal origin: bovine femur and pig ribs [9]. Turkyilmaz et al., Herekar et al. found a significant correlation between Hounsfield units from CT scans and torque and RFA values at implant sites [3, 13]. Tomer at al in their study evaluated a possible correlation between bone densities, insertion torque, and implant stability for total of 30 implants placed in 7 patients [22]. Similarly Mikic M et al. in their study concluded statistically significant strong positive correlation between the bone density expressed in HU units measured in the software of the CBCT device and the primary stability of dental implants expressed in ISQ units. Higher HU values are related to higher ISQ values [8]. Our findings are consistent with these studies and provide additional evidence through the inclusion of photobiomodulation (PBM) as an experimental variable.

A novel contribution of this study is the inclusion of PBM therapy, which was applied using three different protocols. PBM has been shown to stimulate osteoblastic activity, increase vascularization, and accelerate bone remodeling through the modulation of mitochondrial pathways and gene expression related to osteogenesis [15]. Singhania et al. in her study showed improvement in bone density observed post-PBM suggesting a biological enhancement in bone quality, which in turn led to higher insertion torque values even in poor quality (D4) bone [14].

Several in vitro and clinical studies support this mechanism. PBM improves the biochemical environment of the bone-implant interface, reducing inflammatory cytokines while increasing collagen deposition and bone matrix maturation [19, 20, 23]. Matys et al. in his study compared primary and secondary stability using CBCT in D3 bone at baseline and 3 months and found similar results where the value of grayscale at baseline and 12 weeks was significantly higher at the middle and apical level of implants in the laser group in contrast to the control group ($p < 0.01$) [24]. These results were statistically significant in study conducted by Shenoy et al., where PBM positively influenced the tissue healing and improved the ISQ values [18]. In our study, the application of PBM was associated with higher torque values in both D3 and D4 bone types, reinforcing its potential as a pre- and peri-operative adjunct in implant dentistry.

Primary stability, a prerequisite for successful osseointegration, is often quantified via insertion torque and resonance frequency analysis (RFA). While RFA provides data post-placement, insertion torque offers real-time intraoperative feedback and has been validated as a surrogate marker for initial mechanical stability [10, 11]. In low-density bone (e.g., D3, D4), achieving sufficient insertion torque is challenging, and adjunctive strategies such as PBM or bone condensation techniques are often necessary to improve outcomes. Given our findings, incorporating PBM therapy into such a model may further improve predictive accuracy, especially in challenging cases involving low bone density.

Despite the promising results, limitations of this study include the lack of long-term follow-up to assess secondary stability and osseointegration success rates. Future studies should examine whether PBM-induced improvements in primary stability translate into superior long-term implant survival, particularly in high-risk populations.

Conclusion

There is a consistent and statistically significant correlation between CBCT-measured bone density and insertion torque values in both D3 and D4 bone types following PBM therapy. These findings support the use of CBCT as a predictive tool and advocate for PBM as a clinically beneficial adjunct to improve primary implant stability.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

AS, AB, SG, SN, SK were involved in the conception and planning of the study. AS and AB conceived of the presented idea and discussed with all the authors who helped in completing the study. AS, SN, SK carried out the experiment under the supervision of AB and SG. AS drafted the manuscript with the support of SN and SK. AB, SG contributed to overall conceptualisation and design of study. SG, AB served as reviewer after drafting the final manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are included in the published article and the raw data is available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Ethical approval obtained with Institutional Ethics Committee, Datta Meghe Institute of Higher Education and research (DMIHER) with IEC number as DMIMS(DU)/IEC/2021/628 and informed consent taken from patient. Declaration of Helsinki: The study was conducted in accordance with Declaration of Helsinki, an ethics committee approved the protocol and informed consent was obtained from patients.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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